

Raphaël URFIN

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About me

I am a second year PhD student at École Normale Supérieure - PSL in Paris supervised by Giulio Biroli (ENS) and Marc Mézard (Bocconi University) working on the theory of diffusion models. My background is in theoretical physics.

Education

- PhD in Physics and Machine Learning** **ENS-PSL, Paris**
• *Subject: Generative Diffusion and Statistical Physics* **2025 – 2028 (exp.)**
LPENS & Centre de Sciences des Données
Supervisors: Giulio Biroli (ENS) and Marc Mézard (Bocconi University)
- ENS Diploma** **ENS-PSL, Paris**
• *Interdisciplinary Diploma validating my studies at ENS.* **2021 – 2025**
- M2 ICFP Theoretical Physics Track** **ENS-PSL, Paris**
• *First Class Honors, 17.38/20* **2023 – 2024**
Relevant classes: Advanced Statistical Physics, Disordered Systems, Machine Learning.
- M1 ICFP** **ENS-PSL, Paris**
• *First Class Honors, 17.8/20* **2022 – 2023**
Relevant classes: Phase Transitions, Introduction to Quantum Field Theory, General Relativity.
- Bachelor in Physics and Mathematics** **ENS-PSL, Paris**
• *First Class Honors, 17.38/20 (Physics) and 16.36/20 (Mathematics)* **2021 – 2022**

Publications

* denotes equal contribution.

Preprints

- **Raphaël Urfin**, Tony Bonnaire, Giulio Biroli, Marc Mézard.
First Learn, Then Memorize: The Spectral Bias of Diffusion Models.
arXiv:2609.35377 (2026). <https://arxiv.org/abs/2609.35377>
- Jérémie Klinger, **Raphaël Urfin**, Giulio Biroli, Marylou Gabrié.
Weighting Schedules Govern What and When Score-Based Generative Models Learn from Multimodal Data.
arXiv:2609.35322 (2026). <https://arxiv.org/abs/2609.35322>
- **Raphaël Urfin**^{*}, Tony Bonnaire^{*}, Giulio Biroli, Marc Mézard.
Double Descent and Malign Overfitting in Diffusion Models.
arXiv:2609.26392 (2026). <https://arxiv.org/abs/2609.26392>

Conference Proceedings

- Tony Bonnaire^{*}, **Raphaël Urfin**^{*}, Giulio Biroli, Marc Mézard.
Why Diffusion Models Don't Memorize: The Role of Implicit Dynamical Regularization in Training.
NeurIPS 2025, **Best Paper Award (4 papers out of 20,000+ submissions)**
Advances in Neural Information Processing Systems (NeurIPS) 38, 141266-141286
Selected for the JSTAT special issue *Machine Learning 2026: J. Stat. Mech.* (2026) 084007.

Conferences, Workshops and Seminars

Invited Talks.....

- **Generative Memory Lab, Donders Institute** **Radboud University, Netherlands (Online)**
Invited Seminar *September 2026*
- **LemanTH 2026** **Lausanne, Switzerland**
Workshop *April 2026*
- **Vitelli's Group Meeting** **Online**
Group meeting (non-reciprocal p-spins) *March 2026*
- **Scattering Club, Data Science for Astrophysics group @ LPENS** **Online**
Invited Seminar (1 hour) *February 2026*
- **Voxel51 Best of NeurIPS 2025** **Online**
Invited Seminar (30 min) *January 2026*
- **Centre de Sciences des Données Internal Seminar** **Paris, France**
Invited Seminar (30 min) *December 2025*
- **Machine Learning & Signal Processing @ ENS Lyon** **Lyon, France**
Invited Seminar (1 hour) *November 2025*

Contributed Talks.....

- **Journées de Physique Statistique 2026** **Paris, France**
Conference (4-min flash talk) *January 2026*
- **EurIPS 2025** **Copenhagen, Denmark**
Conference (poster + 13-min oral) *December 2025*
- **NeurIPS in Paris 2025** **Paris, France**
Workshop (poster + 10-min oral) *November 2025*
- **StatPhys 29** **Florence, Italy**
Conference (15-min talk) *July 2025*
- **Youth in High Dimensions** **Trieste, Italy**
Workshop (15-min talk) *July 2025*
- **Journées de Physique Statistique 2025** **Paris, France**
Conference (4-min flash talk) *January 2025*

Summer Schools.....

- **Beg Rohu Summer School of Statistical Physics** **Beg Rohu, France**
Summer School *June 2025 & June 2026*
 - **2026 Edition (Optimization and Control of Nonequilibrium Systems):** Lectures by international researchers (e.g. G. Peyré, G. Rotskoff ...). Presented a poster on Memorization/Generalization in diffusion models.
 - **2025 Edition (Learning with Machines, Physics and Mind):** Lectures on Machine Learning and Statistical Physics by international researchers (e.g., Yann LeCun, Julia Kempe, Stéphane Mallat, Marc Mézard ...). Presented a poster on Memorization/Generalization in diffusion models.
- **Complex and Glassy Systems** **Cargèse, France**
Summer School *July 2024*
 - Lectures on Statistical Physics and interdisciplinary applications by international researchers (e.g. Marc Mézard, Eric Vanden-Eijnden, Valentina Ros, Guy Bunin).

Awards and Grants

- 2025: Recipient of one of the four best paper awards at NeurIPS 2025. **(4 papers out of 20,000+ submissions)**
- 2025: CDSN, Competitive PhD scholarship for ENS students

Reviewing

- Machine Learning Workshops: EurIPS 2025 Workshop Principles of Generative Modeling (PriGM), ICLR 2026 Workshop DeLTa, ICLR 2026 Workshop Sci4DL.
- Machine Learning Journals: TMLR

Skills

- Languages: French (Native), English (Fluent), Italian (Fluent).
- Software skills: Python (Pytorch), Matlab, Latex.

Teaching Experience

- TA Phase Transitions (M1 level, Master ICFP, ENS-PSL), 16 hours, taught in English with Giulio Biroli. Fall 2026.
- TA Mathematics Tutoring for first-year students in Physics at ENS, 36 hours, taught in French with Amir-Kian Kashani-Poor. Fall 2025 and Fall 2026.
- TA Machine Learning (M2, Master ICFP, ENS-PSL), 11 hours, taught in English with Marc Lelarge and Leonardo Defilippis. Spring 2026.
- TA Stochastic Processes for Physics (M1 level, Master ICFP, ENS-PSL), 16 hours, taught in English with Marylou Gabri . Fall 2025.
- "Kh lles" (Preparation for the oral exams of French "Grandes  coles") in Mathematics and Physics for Classes Pr paratoires PC, Lyc e Stanislas (2022–2024).

Relevant Experience

- Statistical Physics and Diffusion Models**
Research Internship
– Supervisor: Marc M zard, Department of Computing Sciences.
– Subject: The Memorization/Generalization transition in diffusion models.
– Resulted in a publication accepted at NeurIPS 2025
Bocconi University, Milan, Italy
February – May 2025
- Statistical Physics and Diffusion Models**
Research Internship
– Supervisor: Giulio Biroli, Centre de Sciences des Donn es.
– Subject: The effect of implicit regularization in diffusion models.
ENS-PSL, Paris
April 2024 – January 2025
- Emergent behaviors in large ecosystems**
Research Internship
– Supervisor: Camille Scalliet, Soft Matter Group, Department of Applied Mathematics and Theoretical Physics.
– Using tools from disordered systems (Cavity Method, Random Matrix Theory...) to understand emergent collective behaviors in ecological systems.
University of Cambridge, UK
January – July 2023